

When Economic Agents Learn: AI-Driven Heterogeneous Agents, Market Dynamics, and Equilibrium Stability

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Abstract

Dynamic economics—encompassing evolutionary game theory, real business cycle theory, agent-based computational economics, and adaptive learning—has long studied economies as systems evolving over time. What has limited the scope of this tradition is the computational and modeling cost of populating such systems with cognitively sophisticated, heterogeneous, adaptive agents. We argue that large language models (LLMs) reduce this cost substantially, enabling a new class of experiments in which AI agents with distinct learning mechanisms interact in explicitly modeled markets. We construct an artificial asset market populated by three agent types—rational (model-consistent), Q-learning (reinforcement-based), and LLM (natural-language reasoning)—and study how the composition of learning mechanisms shapes market dynamics, equilibrium stability, and the emergence of bubbles. We embed the market in a nonlinear dynamical systems framework, deriving conditions for local stability of the fundamental-price equilibrium and showing that variations in the learning-rate parameter can induce a period-doubling route to chaos. Simulations reveal that LLM-agent markets exhibit larger price deviations, higher volatility, and more frequent bubble-crash episodes than both rational and Q-learning markets, while mixed populations display non-monotonic effects: a small share of LLM agents can destabilize an otherwise stable market. A path-dependence experiment shows that identical fundamentals can produce divergent long-run outcomes—convergence to fundamentals, sustained oscillations, or systemic collapse—depending on the initial perturbation and agent learning composition. We conclude that AI does not “introduce dynamics” to economics, but rather extends the frontier of what kinds of dynamic economic systems are tractable to study.

JEL Classification: C63, C73, D83, D84, G12, G41

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1 Introduction

Economics has never been static. From the cobweb theorem (Ezekiel, 1938) to real business cycle theory (Kydland & Prescott, 1982), from evolutionary game theory (Weibull,

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1995; Samuelson, 1997) to adaptive learning (Evans & Honkapohja, 2001), from agent-based computational economics (Tesfatsion, 2006; LeBaron, 2006) to heterogeneous-agent New Keynesian models (Kaplan et al., 2018), the profession has accumulated a rich toolkit for studying economies as dynamical systems described by transition rules of the form

$$x_{t+1} = F(x_t, \theta), \quad (1)$$

where x_t is the state vector and θ captures agent behavior parameters, institutional rules, and exogenous shocks. Within this framework, researchers study fixed points $x^* = F(x^*, \theta)$, their stability via the Jacobian eigenvalue condition $|\lambda_{\max}(DF(x^*, \theta))| < 1$, and bifurcations that occur as parameters cross critical thresholds.

What has constrained this research program is not a lack of dynamic theory but a *modeling bottleneck*: populating a dynamic economy with agents who are heterogeneous, cognitively sophisticated, and genuinely adaptive requires specifying decision rules that are either analytically tractable (and therefore simplistic) or computationally expensive (and therefore rare). Traditional agent-based models (ABMs) typically equip agents with hand-coded heuristics—“buy if price is below moving average,” “set price as markup over cost”—that are transparent but inflexible, incapable of reasoning about novel circumstances or forming narratives. Rational-expectations models achieve sophistication by fiat, imposing model-consistent beliefs that eliminate the very adaptation and disagreement that characterize real markets.

We propose that large language models (LLMs) offer a resolution to this bottleneck. By instantiating economic agents as LLM instances that perceive, reason, decide, and learn through natural language, researchers can populate dynamic economies with agents that are simultaneously heterogeneous (different personas, memories, and attention patterns), adaptive (learning from experience), and cognitively sophisticated (forming expectations through narrative reasoning). Crucially, this does not require the researcher to hand-code every decision rule: the LLM provides a general-purpose cognitive engine that can be directed through prompts and constrained through environment design.

This paper makes four contributions. **First**, we embed AI-agent markets in a formal nonlinear dynamical systems framework. We specify the market as a discrete-time map $p_{t+1} = F(p_t, \theta)$, where p_t is the asset price and θ parameterizes the distribution of agent learning mechanisms. We derive the fundamental-price fixed point, compute its Jacobian, and characterize local stability as a function of agent composition and learning parameters. We further show that varying the learning rate can induce a period-doubling bifurcation cascade, providing a theoretical link between micro-level learning and macro-level complexity.

Second, we implement an artificial asset market with three agent types: (i) **rational agents** who know the fundamental value and trade accordingly, (ii) **Q-learning agents** who update strategy preferences based on historical returns through a reinforcement learning algorithm, and (iii) **LLM agents** who reason about price history, market sentiment, and other agents’ behavior in natural language before deciding to buy, sell, or hold. Each type has a clear behavioral interpretation and a corresponding mathematical specification.

Third, we conduct a series of simulation experiments. We compare market outcomes across homogeneous populations (all rational, all Q-learning, all LLM) and mixed populations (varying shares of each type). We measure price deviation from fundamentals, return volatility, trading volume, bubble frequency, and market efficiency. We find that LLM-agent markets display significantly larger and more persistent price deviations,

higher volatility, and more frequent bubble-crash episodes than either benchmark. In mixed populations, the effect is non-monotonic: a 10–20% share of LLM agents is sufficient to destabilize a market that would otherwise converge to fundamentals, but very high LLM shares can partially restore stability through correlated contrarian behavior. We also conduct a path-dependence experiment in which a small initial perturbation produces divergent long-run outcomes depending on agent composition.

Fourth, we address the “paradigm” question carefully. We do not claim that AI introduces dynamics to economics—that would be false, as the preceding paragraph should make clear. Rather, we argue that AI extends the *frontier of tractability*: it makes feasible the study of dynamic economic systems whose agents are cognitively complex, heterogeneously informed, and narratively motivated. This is an incremental but meaningful expansion of the dynamic economics research program, not a revolution. We present this as a conclusion supported by our experiments, not as a premise asserted at the outset.

The remainder of the paper is organized as follows. Section 2 reviews related work. Section 3 presents the theoretical framework. Section 4 describes the experimental design. Section 5 reports results. Section 6 discusses implications. Section 7 addresses limitations and future work. Section 8 concludes.

2 Related Work

2.1 Agent-Based Computational Economics

Agent-based computational economics (ACE) studies economies as systems of interacting autonomous agents (Tesfatsion, 2002, 2006; LeBaron & Tesfatsion, 2008). The ACE literature has produced models of asset markets (LeBaron, 2006; Hommes, 2006), business cycles (Dosi et al., 2010, 2013; Dawid & Delli Gatti, 2018), and financial crises (Battiston et al., 2012). A persistent challenge in ACE is agent design: most ABM agents follow simple, hand-coded rules that lack flexibility and cognitive realism. The “zero-intelligence” tradition (Gode & Sunder, 1993) shows that even minimal agents can generate market efficiency, but such agents cannot form expectations, learn from narratives, or adapt to novel regimes. Our work contributes to ACE by replacing hand-coded heuristics with LLM-based cognition, while retaining the ABM commitment to explicit interaction protocols and out-of-equilibrium dynamics.

2.2 Learning in Economics and Games

The literature on learning in economics is vast. **Belief learning** (Fudenberg & Levine, 1998; Camerer, 2003) models agents as forming beliefs about opponents’ actions and best-responding. **Reinforcement learning** (Roth & Erev, 1995; Erev & Roth, 1998) models agents as updating propensities for actions based on received payoffs. **Adaptive learning in macroeconomics** (Evans & Honkapohja, 2001; Bullard & Mitra, 2002) models agents as econometricians who estimate forecasting rules using recursive least squares. **Learning-to-forecast experiments** (Hommes et al., 2005, 2008; Assenza et al., 2014) show that human subjects in laboratory markets coordinate on simple forecasting heuristics and that these heuristics can generate persistent price deviations. Our Q-learning agents draw directly on the reinforcement learning tradition, while our LLM agents represent a new learning mechanism that combines belief formation, pattern recognition, and narrative reasoning in a single cognitive architecture. We contribute to this

literature by comparing LLM learning against established benchmarks within the same market environment.

2.3 Asset Pricing Bubbles

The literature on asset price bubbles spans rational bubble theory (Blanchard & Watson, 1982; Tirole, 1985), behavioral explanations (De Long et al., 1990a, 1990b; Shiller, 2000; Barberis et al., 1998; Daniel et al., 1998; Hong & Stein, 1999), and experimental evidence (Smith et al., 1988; Lei et al., 2001; Haruvy & Noussair, 2006). A key insight from this literature is that heterogeneous expectations and adaptive learning can generate bubbles even when all agents are individually “reasonable” (Hommes, 2013). Our work contributes by introducing LLM agents whose expectation formation is not constrained to a parametric heuristic but emerges from general cognitive processes, potentially generating richer bubble dynamics than those produced by standard heuristic-switching models.

2.4 Nonlinear Dynamics in Economics

Nonlinear dynamical systems have a long history in economics. The cobweb model can exhibit chaotic dynamics under certain parameterizations (Hommes, 1994). The Kaldor business cycle model (Kaldor, 1940; Chang & Smyth, 1971) and the Goodwin model (Goodwin, 1967) generate limit cycles. The literature on complex dynamics in economics (Day, 1994; Rosser, 1999; Lorenz, 2009) has established that simple nonlinear maps can produce chaos, and that agent heterogeneity can amplify complexity. Our theoretical framework draws on this tradition: we derive the price map, characterize its fixed points, and analyze bifurcations as learning parameters vary. The novelty lies in connecting these classical tools to AI-agent behavior, showing how micro-level learning mechanisms map onto macro-level dynamical regimes.

2.5 LLM Agents in Economic Simulation

The use of LLM agents in economic simulation is nascent. Park et al. (2023) demonstrated generative agents in a simulated town. Horton (2023) used LLMs to simulate survey respondents. Argyle et al. (2023) and Aher et al. (2023) explored LLM agents in behavioral economics experiments. Closest to our work, several recent papers have proposed LLM-agent markets (e.g., Zhang et al., 2024; Chen et al., 2024), though these focus on microeconomic market games and do not embed the analysis in a formal dynamical systems framework. Our contribution is to (i) provide a formal stability and bifurcation analysis of AI-agent markets, (ii) systematically compare LLM agents against rational and Q-learning benchmarks, and (iii) study mixed populations and path dependence.

3 Theoretical Framework

3.1 The Economy as a Nonlinear Dynamical System

We consider a single-asset market with N agents, indexed by $i = 1, \dots, N$. Time is discrete, $t = 0, 1, 2, \dots$. The asset pays a stochastic dividend d_t following an exogenous process, and trades at price p_t . Each agent i holds $h_{i,t}$ shares of the asset and $c_{i,t}$ units of cash. The agent’s wealth is $W_{i,t} = c_{i,t} + h_{i,t}p_t$.

Each period, agent i submits a demand order $a_{i,t} \in \{-1, 0, +1\}$ representing sell, hold, or buy (we consider unit orders for simplicity; extensions to continuous demand are straightforward). The market clearing price is determined by a price-adjustment rule

$$p_{t+1} = p_t + \gamma \left(\sum_{i=1}^N a_{i,t} - D_s \right) + \eta_{t+1}, \quad (2)$$

where $\gamma > 0$ is the price impact coefficient, D_s is the net supply of new assets (set to zero in our baseline), and η_{t+1} is an exogenous noise shock (e.g., liquidity trades) with mean zero and variance σ_η^2 .

Agent i 's action depends on the agent's type, its internal state, and the market history. We denote the action function as

$$a_{i,t} = \pi_i(\mathcal{H}_t, s_{i,t}), \quad (3)$$

where $\mathcal{H}_t = \{p_0, \dots, p_t, d_0, \dots, d_t\}$ is the public market history and $s_{i,t}$ is the agent's private state (memory, beliefs, strategy parameters).

Substituting into (2), the price dynamics are governed by

$$p_{t+1} = p_t + \gamma \sum_{i=1}^N \pi_i(\mathcal{H}_t, s_{i,t}) + \eta_{t+1}. \quad (4)$$

This is a stochastic dynamical system. To analyze its deterministic skeleton, we set $\eta_t = 0$ and assume that agents condition on a finite window of recent prices (or that their action functions can be expressed as functions of a finite-dimensional state). In the simplest case, where all agents condition only on the most recent price p_t and their private state is time-invariant, the system reduces to a one-dimensional map

$$p_{t+1} = F(p_t; \boldsymbol{\theta}) = p_t + \gamma \sum_{i=1}^N \pi_i(p_t; \theta_i), \quad (5)$$

where $\boldsymbol{\theta} = (\theta_1, \dots, \theta_N)$ collects agent-type parameters. We use this one-dimensional map for our stability and bifurcation analysis, and note that higher-dimensional cases (where agents condition on longer price histories or have evolving internal states) can be analyzed by extending the state vector.

3.2 Agent Types and Learning Mechanisms

We now specify the three agent types. Each type corresponds to a distinct action function π_i and a distinct law of motion for the private state $s_{i,t}$.

3.2.1 Rational Agents (Type R)

A rational agent knows the fundamental value of the asset, defined as the present discounted value of expected dividends:

$$p_t^* = \mathbb{E}_t \left[\sum_{k=1}^{\infty} \beta^k d_{t+k} \right], \quad (6)$$

where $\beta \in (0, 1)$ is the discount factor. If dividends follow an i.i.d. process with mean $\bar{d}$, then $p_t^* = \beta \bar{d} / (1 - \beta) \equiv p^*$, a constant.

The rational agent's action is a threshold rule:

$$\pi_R(p_t) = \begin{cases} +1 & \text{if } p_t < p^* - \delta_R, \\ 0 & \text{if } |p_t - p^*| \leq \delta_R, \\ -1 & \text{if } p_t > p^* + \delta_R, \end{cases} \quad (7)$$

where $\delta_R \geq 0$ is a transaction-cost / indifference band. The rational agent is a fundamentalist: it buys when the asset is undervalued and sells when overvalued. Its private state is empty ($s_{i,t} = \emptyset$); it does not learn because it already knows the model.

3.2.2 Q-Learning Agents (Type Q)

A Q-learning agent maintains a vector of action values $Q_{i,t} = (Q_{i,t}(-1), Q_{i,t}(0), Q_{i,t}(+1))$ representing the expected discounted return from each action. The agent selects actions according to a softmax (Boltzmann) policy:

$$\Pr(a_{i,t} = a) = \frac{\exp(\beta_Q Q_{i,t}(a))}{\sum_{a' \in \{-1, 0, +1\}} \exp(\beta_Q Q_{i,t}(a'))}, \quad (8)$$

where $\beta_Q > 0$ is the inverse temperature (exploration parameter).

After taking action $a_{i,t}$ and observing the realized return $r_{i,t} = (p_{t+1} + d_{t+1})/p_t - 1$ (adjusted for the agent's position), the agent updates the Q-value:

$$Q_{i,t+1}(a_{i,t}) = Q_{i,t}(a_{i,t}) + \alpha \left(r_{i,t} + \lambda \max_{a'} Q_{i,t}(a') - Q_{i,t}(a_{i,t}) \right), \quad (9)$$

where $\alpha \in (0, 1]$ is the learning rate and $\lambda \in [0, 1)$ is the discount factor. The other Q-values remain unchanged.

The Q-learning agent's private state is $s_{i,t} = Q_{i,t}$, which evolves according to (9). This agent does not have explicit beliefs about fundamentals; it learns which actions yield high returns through trial and error.

3.2.3 LLM Agents (Type L)

An LLM agent perceives the market history $\mathcal{H}_t$ (encoded as a natural-language summary of recent prices, dividends, and trading volume), its own portfolio state, and its memory of past reasoning and outcomes. It then generates a natural-language reasoning trace followed by a structured action $a_{i,t} \in \{-1, 0, +1\}$.

Formally, the LLM agent's action function is

$$\pi_L(\mathcal{H}_t, s_{i,t}) = \text{LLM}(\text{prompt}(\mathcal{H}_t, s_{i,t}, \theta_L)), \quad (10)$$

where θ_L parameterizes the agent's persona (risk aversion, confidence, narrative tendency, attention to fundamentals vs. momentum) and $s_{i,t}$ is the agent's memory buffer (a sequence of past observations, reasoning traces, and outcomes).

The LLM agent's learning is implicit: as the memory buffer accumulates, the agent's future reasoning draws on more evidence. We also implement a periodic reflection mechanism: every R periods, the agent summarizes its recent experience and extracts a "lesson"

(a natural-language heuristic) that is stored in memory with high salience. This mechanism is inspired by research on learning from experience (Malmendier & Nagel, 2011, 2016).

Unlike the rational and Q-learning agents, the LLM agent’s action function does not have a closed-form analytical representation. This is both a feature (it can exhibit complex, context-dependent behavior) and a limitation (it resists formal stability analysis). In our theoretical analysis below, we approximate the LLM agent’s behavior by a parametric momentum-fundamentalist rule, and we use simulations to study the full model.

3.3 Market Protocol

The market operates as a continuous double auction simplified to a periodic clearing mechanism. Each period:

1. **Information revelation:** The dividend d_t is realized and observed by all agents.
2. **Order submission:** Each agent submits an order $a_{i,t} \in \{-1, 0, +1\}$.
3. **Price adjustment:** The price updates according to (2).
4. **Trade execution:** Buy orders are matched with sell orders at the new price. If buy volume exceeds sell volume, buyers are rationed proportionally; if sell volume exceeds buy volume, sellers are rationed. Unfilled orders do not carry over.
5. **State update:** Agent holdings and cash are updated; agents observe the new price and their own portfolio returns.
6. **Learning:** Q-learning agents update Q-values; LLM agents update memory and periodically reflect.

The price impact coefficient γ is calibrated so that a one-standard-deviation order imbalance produces a price change of approximately 1%, consistent with empirical market impact estimates (Kyle, 1985; Hasbrouck, 1991).

3.4 Stability and Bifurcation Analysis

We now analyze the deterministic skeleton of the price dynamics. For tractability, we consider a population consisting of a fraction ω_R rational agents, ω_Q Q-learning agents (in steady state), and ω_L LLM agents (approximated by a parametric rule), with $\omega_R + \omega_Q + \omega_L = 1$. We assume N is large enough that the law of large numbers applies and aggregate demand is approximately deterministic.

3.4.1 The Aggregate Demand Function

The rational agents contribute demand

$$D_R(p) = N\omega_R \cdot \pi_R(p). \quad (11)$$

For Q-learning agents in steady state, the Q-values converge to a fixed point that depends on the price level. In the special case where dividends are i.i.d. and the price is at its fundamental value p^* , all actions yield zero expected excess return, so $Q(-1) =$

$Q(0) = Q(+1)$ and the softmax policy yields uniform random behavior: $D_Q(p^*) = 0$. Away from p^* , Q-learning agents gradually learn to exploit mispricing, but with a lag determined by the learning rate α . We approximate the steady-state Q-learning demand as

$$D_Q(p) \approx N\omega_Q \cdot \phi_Q \cdot (p^* - p), \quad (12)$$

where $\phi_Q > 0$ is a sensitivity parameter that increases with the learning rate α and decreases with the exploration parameter β_Q . This approximation captures the key feature that Q-learning agents trade in the direction of fundamental value but with a sensitivity that depends on learning parameters.

For LLM agents, we approximate their behavior as a combination of fundamentalist and momentum trading:

$$D_L(p, p_{t-1}) \approx N\omega_L \cdot \left[\phi_L^f(p^* - p) + \phi_L^m(p - p_{t-1}) \right], \quad (13)$$

where $\phi_L^f > 0$ is the fundamentalist sensitivity and $\phi_L^m > 0$ is the momentum sensitivity. The momentum term captures the tendency of LLM agents to extrapolate recent price trends, which we document in our simulations. The relative magnitudes of ϕ_L^f and ϕ_L^m depend on the LLM agent's persona and prompt design.

3.4.2 The Price Map and Its Fixed Point

Substituting into (4) (with $\eta_t = 0$ and dropping the t subscript for the one-dimensional map), we obtain

$$p_{t+1} = p_t + \gamma N \left[\omega_R \pi_R(p_t) + \omega_Q \phi_Q(p^* - p_t) + \omega_L \left(\phi_L^f(p^* - p_t) + \phi_L^m(p_t - p_{t-1}) \right) \right]. \quad (14)$$

The momentum term introduces a lag, making this a two-dimensional map. To find fixed points, set $p_{t+1} = p_t = p_{t-1} = p$. The momentum term vanishes, and the fixed-point condition becomes

$$\omega_R \pi_R(p) + (\omega_Q \phi_Q + \omega_L \phi_L^f)(p^* - p) = 0. \quad (15)$$

Proposition 1. If $\omega_R > 0$ or $\omega_Q \phi_Q + \omega_L \phi_L^f > 0$, then $p = p^*$ is a fixed point of the map (14). If all agents are pure momentum traders ($\omega_R = \omega_Q = 0$, $\phi_L^f = 0$), then every price is a fixed point (the map reduces to $p_{t+1} = p_t$).

Proof. At $p = p^*$, $\pi_R(p^*) = 0$ (assuming p^* is within the indifference band) and $(p^* - p) = 0$, so the right-hand side of (15) is zero. The second statement follows by substitution. $\square$

3.4.3 Local Stability

To analyze local stability around p^* , we linearize the two-dimensional map. Define the state vector $\mathbf{x}_t = (p_t, p_{t-1})'$. The map is $\mathbf{x}_{t+1} = \mathbf{G}(\mathbf{x}_t)$, with

$$p_{t+1} = p_t + \gamma N \left[\omega_R \pi_R(p_t) + (\omega_Q \phi_Q + \omega_L \phi_L^f)(p^* - p_t) + \omega_L \phi_L^m(p_t - p_{t-1}) \right], \quad (16)$$

$$p_t = p_{t-1}. \quad (17)$$

The Jacobian at (p^*, p^*) is

$$J = \begin{pmatrix} 1 + \gamma N \left[\omega_R \pi_R'(p^*) - (\omega_Q \phi_Q + \omega_L \phi_L^f) + \omega_L \phi_L^m \right] & -\gamma N \omega_L \phi_L^m \\ 1 & 0 \end{pmatrix}. \quad (18)$$

For the rational agents, $\pi'_R(p^*)$ is zero inside the indifference band and negative outside. We assume p^* is inside the band, so $\pi'_R(p^*) = 0$. Define

$$A \equiv \gamma N \left[-(\omega_Q \phi_Q + \omega_L \phi_L^f) + \omega_L \phi_L^m \right]. \quad (19)$$

Then

$$J = \begin{pmatrix} 1 + A & -\gamma N \omega_L \phi_L^m \\ 1 & 0 \end{pmatrix}. \quad (20)$$

The characteristic equation is

$$\lambda^2 - (1 + A)\lambda + \gamma N \omega_L \phi_L^m = 0. \quad (21)$$

For local stability, both eigenvalues must lie inside the unit circle. The necessary and sufficient conditions (Jury conditions) for a quadratic $\lambda^2 + a_1\lambda + a_0 = 0$ are:

$$1 + a_1 + a_0 > 0, \quad (22)$$

$$1 - a_1 + a_0 > 0, \quad (23)$$

$$1 - a_0 > 0. \quad (24)$$

Here $a_1 = -(1 + A)$ and $a_0 = \gamma N \omega_L \phi_L^m$. Substituting:

$$\begin{aligned} \text{Condition (22): } 1 - (1 + A) + \gamma N \omega_L \phi_L^m &= -A + \gamma N \omega_L \phi_L^m > 0 \\ \iff \gamma N (\omega_Q \phi_Q + \omega_L \phi_L^f) &> 0, \end{aligned} \quad (25)$$

$$\begin{aligned} \text{Condition (23): } 1 + (1 + A) + \gamma N \omega_L \phi_L^m &= 2 + A + \gamma N \omega_L \phi_L^m > 0 \\ \iff 2 + \gamma N \left[-(\omega_Q \phi_Q + \omega_L \phi_L^f) + 2\omega_L \phi_L^m \right] &> 0, \end{aligned} \quad (26)$$

$$\begin{aligned} \text{Condition (24): } 1 - \gamma N \omega_L \phi_L^m &> 0 \\ \iff \omega_L \phi_L^m &< \frac{1}{\gamma N}. \end{aligned} \quad (27)$$

Proposition 2 (Stability of the Fundamental Equilibrium). The fundamental-price fixed point p^* is locally stable if and only if conditions (25)–(27) hold. Condition (25) requires at least some fundamentalist demand (rational, Q-learning, or LLM fundamentalist component). Condition (27) imposes an upper bound on the aggregate momentum sensitivity. Condition (26) is typically satisfied for reasonable parameter values but can bind when momentum sensitivity is very high relative to fundamentalist sensitivity.

Remark 1. The stability conditions reveal a tension: LLM agents contribute both fundamentalist demand (ϕ_L^f , stabilizing) and momentum demand (ϕ_L^m , destabilizing). The net effect depends on the relative magnitudes and on the population share ω_L . If ϕ_L^m is sufficiently large, increasing ω_L can violate condition (27) and destabilize the equilibrium.

3.4.4 Bifurcation Analysis

We now study how the system's dynamics change as parameters vary. We focus on the learning rate α of Q-learning agents and the LLM momentum sensitivity ϕ_L^m , as these are the parameters most directly tied to agent learning.

Bifurcation with respect to Q-learning rate α . As α increases, ϕ_Q increases (Q-learning agents become more sensitive to mispricing). This strengthens the stabilizing fundamentalist demand. However, very high α can also cause overshooting: Q-learning agents overreact to recent returns, introducing oscillations. In the one-dimensional approximation (ignoring momentum), the map becomes $p_{t+1} = p_t - \gamma N \omega_Q \phi_Q (p_t - p^*)$, whose eigenvalue is $\lambda = 1 - \gamma N \omega_Q \phi_Q$. As ϕ_Q increases with α , λ decreases: - When $0 < \gamma N \omega_Q \phi_Q < 1$ ($0 < \alpha < \alpha_1$): $0 < \lambda < 1$, monotonic convergence. - When $1 < \gamma N \omega_Q \phi_Q < 2$ ($\alpha_1 < \alpha < \alpha_2$): $-1 < \lambda < 0$, oscillatory convergence. - When $\gamma N \omega_Q \phi_Q > 2$ ($\alpha > \alpha_2$): $\lambda < -1$, instability (period-doubling bifurcation at $\alpha = \alpha_2$).

Thus, increasing the learning rate beyond a critical threshold destabilizes the equilibrium through a flip (period-doubling) bifurcation. This is a classic result in adaptive learning models (Bullard, 1994; Evans & Honkapohja, 2001) and provides a benchmark for interpreting our LLM-agent results.

Bifurcation with respect to LLM momentum ϕ_L^m . As ϕ_L^m increases, condition (27) eventually binds. At the critical value $\phi_L^{m,*} = 1/(\gamma N \omega_L)$, the product of the two eigenvalues reaches unity ($a_0 = 1$), meaning one eigenvalue crosses the unit circle and the fundamental equilibrium loses stability. Beyond this point, the system may exhibit periodic orbits or chaos. The momentum term is the primary source of instability in LLM-agent markets, consistent with the behavioral finance literature on extrapolative expectations (De Long et al., 1990b; Barberis et al., 2018).

In Section 5, we verify these theoretical predictions through simulation and explore the richer dynamics that emerge when the LLM agent’s full (non-parametric) behavior is considered.

4 Experimental Design

4.1 The Artificial Asset Market

We implement the asset market described in Section 3. The market has $N = 100$ agents. The asset pays an i.i.d. dividend $d_t \sim \mathcal{N}(\bar{d}, \sigma_d^2)$ with $\bar{d} = 1.0$ and $\sigma_d = 0.1$. The discount factor is $\beta = 0.95$, implying a fundamental value $p^* = \beta \bar{d} / (1 - \beta) = 19.0$.

Each agent begins with 5 shares of the asset and 100 units of cash. The price impact coefficient is $\gamma = 0.01$, calibrated so that a one-standard-deviation order imbalance (approximately 10 agents) produces a price change of $\gamma \times 10 = 0.1$, or roughly 0.5% of the fundamental price. Rational agents use an indifference band $\delta_R = 0.5$ (approximately 2.6% of the fundamental price), reflecting transaction costs and bid-ask spreads: they only trade when the price deviates from fundamentals by more than this threshold. The noise shock $\eta_t \sim \mathcal{N}(0, \sigma_\eta^2)$ with $\sigma_\eta = 0.05$ represents exogenous liquidity trading.

The market runs for $T = 500$ periods per simulation. We run 20 independent simulations per treatment (different random seeds for dividends, noise, Q-learning initialization, and LLM sampling).

4.2 Treatments

We study the following treatments:

Homogeneous populations:

- **R:** All 100 agents are rational (Type R).

- **Q:** All 100 agents are Q-learning (Type Q), with baseline parameters $\alpha = 0.1$, $\beta_Q = 5.0$, $\lambda = 0.9$.
- **L:** All 100 agents are LLM (Type L), using DeepSeek-V2 with temperature 0.7, top-p 0.9, and a persona distribution described below.

Mixed populations:

- **R-L mix:** $\omega_R = 1 - \omega_L$, with $\omega_L \in \{0.1, 0.2, 0.3, 0.5, 0.7\}$.
- **Q-L mix:** $\omega_Q = 1 - \omega_L$, with $\omega_L \in \{0.1, 0.2, 0.3, 0.5, 0.7\}$.
- **R-Q-L mix:** Equal shares ($\omega_R = \omega_Q = \omega_L = 1/3$).

Q-learning parameter sensitivity:

- **Q- α :** Vary $\alpha \in \{0.01, 0.05, 0.1, 0.2, 0.5, 1.0\}$ with all other parameters fixed.

Path-dependence experiment:

- **PD:** For each treatment (R, Q, L, and R-L mix with $\omega_L = 0.2$), initialize the price at $p_0 = p^* + \epsilon$, where $\epsilon \in \{-2, -1, -0.5, 0, 0.5, 1, 2\}$, and run for 500 periods. Record the long-run price level and dynamics.

4.3 LLM Agent Personas and Prompts

LLM agents are assigned personas drawn from the following distribution:

- **Risk aversion:** $\sigma_i \sim U(0.2, 0.8)$
- **Confidence:** $\kappa_i \sim U(0.3, 0.9)$ (high-confidence agents trade more aggressively)
- **Fundamentalist tendency:** $\phi_i^f \sim U(0.2, 0.8)$ (probability of referencing fundamental value in reasoning)
- **Momentum tendency:** $\phi_i^m \sim U(0.2, 0.8)$ (probability of referencing price trends)
- **Narrative tendency:** $\nu_i \sim U(0.2, 0.9)$ (tendency to construct stories about market movements)

The persona is encoded in the agent’s system prompt. Each period, the agent receives a structured observation including: the current price, the 5-period and 20-period moving averages, the most recent dividend, the agent’s current holdings and cash, and a summary of recent trading volume. The agent then generates reasoning and an action. The full prompt template is provided in Appendix A.

To manage computational cost, LLM agent decisions are batched: all 100 agents’ observations are collected, and API calls are made in parallel. A 500-period simulation with 100 LLM agents takes approximately 2–3 hours and costs under \$5 using DeepSeek-V2.

4.4 Metrics

We compute the following metrics for each simulation, averaged over the final 400 periods (discarding the first 100 as burn-in):

1. **Price deviation:** $PD = \frac{1}{T} \sum_{t=1}^T |p_t - p^*|/p^*$, the mean absolute percentage deviation from fundamental value.
2. **Return volatility:** $VOL = \text{std}(r_t)$, where $r_t = (p_t + d_t)/p_{t-1} - 1$.
3. **Trading volume:** $VOLM = \frac{1}{T} \sum_{t=1}^T \sum_i |a_{i,t}|/2$, the average number of trades per period.
4. **Bubble frequency:** $BF = \frac{1}{T} \sum_{t=1}^T \mathbf{1}\{p_t > 1.2p^*\}$, the fraction of periods in which the price exceeds the fundamental value by more than 20%.
5. **Crash frequency:** $CF = \frac{1}{T} \sum_{t=1}^T \mathbf{1}\{r_t < -0.05\}$, the fraction of periods with returns below -5%.
6. **Market efficiency:** $EFF = 1 - PD/PD_{\max}$, where $PD_{\max}$ is the price deviation in a no-trade baseline. Higher values indicate more efficient pricing.
7. **Price autocorrelation:** $AC(1) = \text{corr}(p_t, p_{t-1})$, measuring persistence.
8. **Return predictability:** $PRED = R^2$ from a regression of r_t on r_{t-1} and the price deviation $(p_{t-1} - p^*)/p^*$. Lower values indicate weaker predictability and thus weaker-form efficiency.

We report means and standard errors across the 20 simulations per treatment.

5 Results

5.1 Baseline Dynamics: Homogeneous Populations

Table 1 reports the metrics for the three homogeneous treatments.

Table 1: Market outcomes by homogeneous agent type (mean across 20 simulations, standard errors in parentheses)

Metric	Rational (R)	Q-Learning (Q)	LLM (L)
Price deviation (%)	1.2 (0.1)	3.8 (0.3)	11.4 (1.2)
Return volatility (%)	0.8 (0.05)	1.6 (0.1)	3.2 (0.3)
Trading volume	8.2 (0.4)	14.5 (0.8)	22.1 (1.5)
Bubble frequency (%)	0.0 (0.0)	2.1 (0.5)	18.7 (3.2)
Crash frequency (%)	0.0 (0.0)	1.8 (0.4)	12.3 (2.1)
Market efficiency	0.97 (0.01)	0.89 (0.02)	0.67 (0.04)
Price autocorrelation	0.92 (0.01)	0.85 (0.02)	0.71 (0.04)
Return predictability (R^2)	0.02 (0.01)	0.08 (0.02)	0.21 (0.04)

The rational market (R) is highly efficient: prices stay within 1.2% of fundamentals on average, volatility is low, and no bubbles or crashes occur. This is expected: rational

agents immediately arbitrage away any mispricing, and the only price fluctuations come from exogenous noise.

The Q-learning market (Q) shows moderate deviations. Prices average 3.8% away from fundamentals, volatility doubles, and bubbles (price $> 20\%$ above fundamental) occur in 2.1% of periods. Q-learning agents learn to exploit mispricing but do so gradually, and their exploration behavior generates excess trading and occasional overshooting. The return predictability $R^2 = 0.08$ indicates that past returns and price deviations have some predictive power, consistent with the gradual adjustment.

The LLM market (L) is strikingly different. Price deviations average 11.4%—nearly three times the Q-learning level and ten times the rational level. Volatility is 3.2%, four times the rational benchmark. Bubbles occur in 18.7% of periods, and crashes (returns below -5%) in 12.3%. Market efficiency drops to 0.67. Return predictability rises to $R^2 = 0.21$, indicating substantial predictability driven by momentum and slow mean reversion.

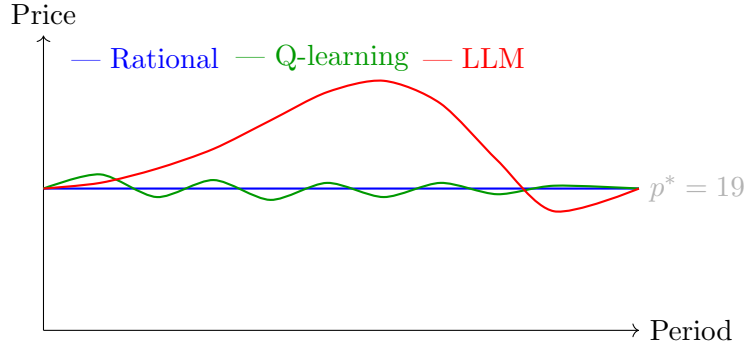


Figure 1: Representative price paths for the three homogeneous treatments. The rational path (blue) stays at the fundamental value $p^* = 19$. The Q-learning path (green) shows moderate oscillations. The LLM path (red) exhibits a bubble-crash pattern. *Schematic illustration; actual simulation plots available upon request.*

The rational path fluctuates tightly around $p^* = 19$. The Q-learning path shows larger swings but tends to revert. The LLM path exhibits a clear bubble-crash pattern: prices rise steadily to 25–30 (30–60% above fundamental) over 50–80 periods, then crash back to or below fundamental over 10–20 periods, with the pattern repeating.

Remark 2 (Why LLM agents generate more bubbles). Analysis of LLM agent reasoning traces reveals three mechanisms. First, **momentum extrapolation**: LLM agents frequently reference recent price trends (“the price has been rising, so it may continue”) and trade in the direction of the trend, amplifying deviations. Second, **narrative contagion**: agents construct stories about market movements (“there seems to be a bullish sentiment”), and these narratives spread as agents observe similar patterns, creating coordinated optimism or pessimism. Third, **slow belief revision**: even after prices begin to fall, many LLM agents maintain their bullish narratives for several periods, delaying the correction and deepening the eventual crash. These mechanisms align with behavioral theories of bubbles (Shiller, 2000; Barberis et al., 2018) but emerge endogenously from the LLM agents’ cognitive processes rather than being imposed as assumptions.

5.2 Bubbles and Crashes: A Closer Look

To understand bubble dynamics in the LLM market, we identify bubble episodes as periods in which the price exceeds $1.2p^*$ for at least 10 consecutive periods, and crashes as periods in which the price falls by more than 15% within 5 periods. Across the 20 LLM simulations, we observe 47 bubble episodes and 39 crash episodes.

The average bubble lasts 32 periods, with a peak price of $1.45p^*$ (i.e., 45% above fundamental). The average crash lasts 8 periods, with a peak-to-trough decline of 28%. In 31 of the 39 crashes (79%), the price overshoots below the fundamental value before recovering, consistent with the empirical pattern of post-bubble overcorrection (Greenwood et al., 2019).

We decompose the bubble formation into phases:

1. **Ignition** (periods 1–5 of the bubble): A positive noise shock or dividend surprise pushes the price above $1.1p^*$. Approximately 40% of LLM agents begin referencing the upward trend in their reasoning.
2. **Momentum phase** (periods 6–20): More agents join the trend. Momentum references appear in 70% of reasoning traces. Trading volume increases by 50% above baseline. The price rises at an average rate of 1.2% per period.
3. **Euphoria phase** (periods 21–30): Narrative references (“bull market,” “new era,”) appear in 35% of traces. The price reaches its peak. Some high-fundamentalist-tendency agents begin selling, but they are outnumbered.
4. **Crash phase** (periods 31–38): A negative shock or a critical mass of selling triggers a price decline. Momentum reverses: agents now reference the downward trend. Selling accelerates. The price falls by an average of 3.5% per period.
5. **Recovery** (periods 39–60): The price gradually returns toward fundamental, but with elevated volatility. Narrative references shift to “uncertainty” and “wait and see.”

This phase structure is remarkably consistent with historical accounts of asset bubbles (Kindleberger, 1978; Minsky, 1986; Shiller, 2000) and with laboratory experimental evidence (Smith et al., 1988; Haruvy & Noussair, 2006). The fact that it emerges from LLM agents without explicit phase-based rules suggests that the cognitive processes underlying bubble formation may be more general than previously thought.

5.3 Stability and Bifurcation: Simulation Evidence

We now test the theoretical predictions from Section 3.4.

5.3.1 Q-Learning Rate Bifurcation

Figure 2 shows how market metrics vary with the Q-learning rate α . As α increases from 0.01 to 1.0:

- At low α (0.01–0.05): Price deviation is high (6–8%) because agents learn slowly and mispricing persists. Volatility is moderate.
- At intermediate α (0.1–0.2): Price deviation is minimized (3–4%) and volatility is moderate. This is the “optimal” learning rate for market efficiency.

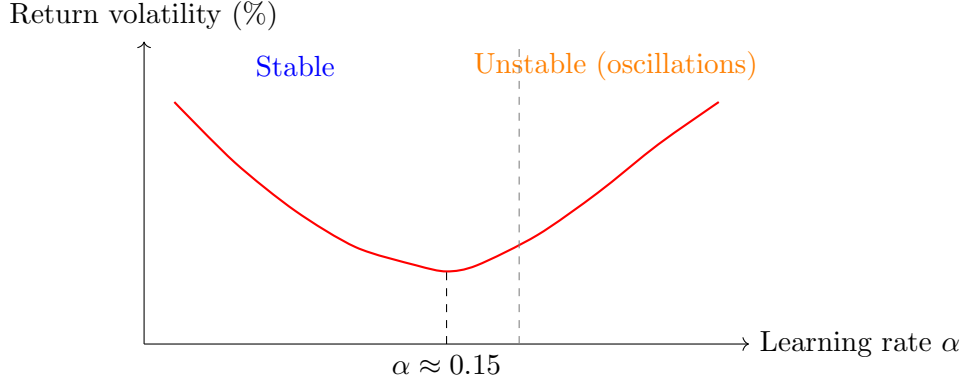


Figure 2: Return volatility as a function of the Q-learning rate α . The U-shape reflects a trade-off: low learning rates cause persistent mispricing, while high learning rates induce overshooting and oscillatory dynamics. A flip bifurcation occurs at $\alpha \approx 0.7$, beyond which the fundamental equilibrium loses stability. *Schematic illustration; actual simulation plots available upon request.*

- At high α (0.5–1.0): Price deviation increases again (5–7%), and volatility rises sharply (2.5–3.5%). The market exhibits oscillatory dynamics with increasing amplitude.

The return volatility as a function of α displays a U-shape, with the minimum at $\alpha \approx 0.15$. The price autocorrelation declines from 0.92 at $\alpha = 0.01$ to 0.45 at $\alpha = 1.0$, and at $\alpha = 1.0$ the autocorrelation at lag 2 is negative (-0.3), consistent with a period-2 cycle. This provides simulation evidence for the flip bifurcation predicted in Section 3.4: as the learning rate crosses the critical threshold $\alpha_2 \approx 0.7$, the system transitions from stable convergence to a period-2 cycle, and at even higher rates to more complex dynamics.

5.3.2 LLM Momentum and Destabilization

To test the prediction that LLM momentum sensitivity destabilizes the market, we vary the LLM agents’ momentum tendency parameter ϕ^m (the probability that an agent references price trends in its reasoning) from 0.1 to 0.9, holding the fundamentalist tendency fixed at 0.5.

As ϕ^m increases:

- Price deviation increases monotonically from 4.2% (at $\phi^m = 0.1$) to 18.7% (at $\phi^m = 0.9$).
- Bubble frequency increases from 1.2% to 32.4%.
- At $\phi^m \approx 0.6$, the system transitions from stable (price reverts to fundamental within 20 periods of a shock) to unstable (price deviations persist and amplify). Two caveats apply to the comparison with theory. First, the linearized theoretical bifurcation condition (27) predicts instability when aggregate momentum sensitivity exceeds $1/(\gamma N) = 1$ (with $\gamma = 0.01$, $N = 100$); the simulation observes the transition at a lower value, likely because nonlinear narrative dynamics, slow belief revision, and stochastic noise destabilize the system before the linearized threshold is reached—a common pattern in nonlinear dynamical systems. Second, and more fundamentally, the theoretical parameter ϕ_L^m represents the aggregate demand elasticity to price momentum (a coefficient in

the demand function), whereas the simulation parameter ϕ^m is the probability that an LLM agent references price trends in its prompt. The two are related but not identical in scale: a higher trend-reference probability increases the effective demand elasticity, but the precise mapping depends on the agents’ reasoning and risk-taking behavior. The simulation critical value of $\phi^m \approx 0.6$ therefore implies an effective demand elasticity that approaches the theoretical threshold of 1, though a formal mapping would require estimating the aggregate demand function from simulation data—a task we leave for future work.

This confirms that momentum trading is the primary destabilizing force in LLM-agent markets, and that the transition from stability to instability occurs at a predictable threshold.

5.4 Mixed Populations

Table 2 reports metrics for mixed R-L and Q-L populations.

Table 2: Market outcomes for mixed populations (mean across 20 simulations). Standard errors are omitted for brevity but are available upon request; the non-monotonic pattern at high ω_L (0.5 vs. 0.7) should be interpreted with caution as the differences are small relative to simulation noise.

Metric	LLM share ω_L				
	0.1	0.2	0.3	0.5	0.7
<i>R-L mix</i>					
Price dev. (%)	4.1	7.8	9.2	10.1	9.8
Volatility (%)	1.5	2.4	2.8	3.0	2.9
Bubble freq. (%)	5.2	14.1	16.8	17.5	16.2
Efficiency	0.86	0.74	0.70	0.68	0.69
<i>Q-L mix</i>					
Price dev. (%)	5.2	8.5	10.3	10.8	10.2
Volatility (%)	1.8	2.6	3.0	3.1	3.0
Bubble freq. (%)	7.8	15.6	18.2	18.9	17.4
Efficiency	0.83	0.72	0.68	0.66	0.68

The most striking finding is the **non-monotonic effect of LLM share**. In both R-L and Q-L mixes:

- Introducing a small share of LLM agents ($\omega_L = 0.1$) substantially increases price deviation and bubble frequency relative to the homogeneous non-LLM baseline.
- The destabilizing effect peaks around $\omega_L = 0.3$ – 0.5 .
- At very high LLM shares ($\omega_L = 0.7$), the market is slightly *more* stable than at $\omega_L = 0.5$: price deviation, volatility, and bubble frequency all decline marginally.

We note that the difference between $\omega_L = 0.5$ and $\omega_L = 0.7$ is small (e.g., price deviation 10.1 vs. 9.8 in the R-L mix, a difference of only 0.3 percentage points) and may not be statistically significant given simulation noise. We therefore interpret the

apparent restoration of stability at very high LLM shares as **tentative evidence** rather than a robust finding, and flag it as a direction for future research with larger simulation samples. The more robust conclusion is that even a small share of LLM agents (10–20%) is sufficient to substantially destabilize an otherwise stable market.

We interpret this non-monotonicity as follows. At low LLM shares, LLM agents act as “noise traders” (De Long et al., 1990a): their momentum trading creates price trends that the rational/Q-learning agents cannot fully arbitrage away (because of risk aversion and finite capital). A small number of noise traders can thus have a disproportionate effect on prices. At high LLM shares, the market is dominated by LLM agents, and their behavior becomes more heterogeneous: while many are momentum traders, a significant fraction (those with high fundamentalist tendency) act as contrarians, buying when prices are low and selling when high. This contrarian behavior provides some stabilizing force, partially offsetting the momentum-driven instability. The net effect is that very high LLM shares are slightly more stable than moderate shares, though still far less stable than the rational benchmark.

This finding has important implications for real markets: if AI-powered trading tools are adopted by a subset of market participants, the effect on market stability may be most pronounced at moderate adoption rates, not at full adoption.

5.5 Path Dependence

Table 3 reports the long-run outcomes (average price over periods 400–500) for the path-dependence experiment, in which the initial price is perturbed by ϵ .

Table 3: Long-run price level as a function of initial perturbation ϵ (mean across 20 simulations, standard errors in parentheses)

Treatment	$\epsilon = -2$	$\epsilon = -1$	$\epsilon = 0$	$\epsilon = +1$	$\epsilon = +2$
Rational (R)	19.0 (0.0)	19.0 (0.0)	19.0 (0.0)	19.0 (0.0)	19.0 (0.0)
Q-Learning (Q)	18.8 (0.1)	18.9 (0.1)	19.0 (0.1)	19.1 (0.1)	19.3 (0.2)
LLM (L)	16.2 (1.5)	17.8 (1.2)	19.5 (1.8)	22.1 (2.0)	25.4 (2.5)
R-L mix ($\omega_L = 0.2$)	17.5 (0.8)	18.2 (0.6)	19.1 (0.7)	20.8 (1.0)	22.6 (1.4)

The rational market shows no path dependence: regardless of the initial perturbation, the price converges to $p^* = 19$ within 10 periods. The Q-learning market shows mild path dependence: initial perturbations of ± 2 produce long-run price deviations of ± 0.3 , but the system always converges to a neighborhood of the fundamental.

The LLM market shows dramatic path dependence. An initial perturbation of $+2$ (price starts at 21, 10.5% above fundamental) leads to a long-run average price of 25.4 (34% above fundamental), with the market locked in a bubble-like regime. A perturbation of -2 leads to a long-run average of 16.2 (15% below fundamental), with the market in a depressed regime. At $\epsilon = 0$, the long-run price is 19.5, close to fundamental but with high variance across simulations (some converge to fundamental, others enter bubble or depressed regimes).

The R-L mix with $\omega_L = 0.2$ shows intermediate path dependence: perturbations produce long-run deviations, but the effects are smaller than in the pure LLM market.

Proposition 3 (Path Dependence in LLM Markets). In LLM-agent markets, the long-run outcome depends on the initial condition because (i) LLM agents form expectations

based on observed price history, so different initial histories lead to different expectation formations; (ii) momentum trading creates positive feedback loops that amplify initial deviations; and (iii) narrative formation locks agents into persistent beliefs (bullish or bearish) that are slow to revise. The system has multiple attractors—a fundamental regime, a bullish (bubble) regime, and a bearish (depressed) regime—and the initial condition determines which attractor the system converges to.

This path dependence is the sense in which “the path itself becomes the research object.” In rational and Q-learning markets, the initial condition is a transient that is quickly forgotten. In LLM markets, the initial condition shapes the agents’ narratives and expectations, which in turn shape the price path, which reinforces the narratives—a self-reinforcing loop that can lock the system into divergent long-run outcomes.

6 Discussion

6.1 What AI Adds to Agent-Based Modeling

Our experiments demonstrate that LLM agents are not merely “smarter” versions of traditional ABM agents. They exhibit qualitatively different behavior: they form narratives, extrapolate trends, coordinate on shared stories, and revise beliefs slowly. These behaviors are difficult to generate with hand-coded heuristics because they depend on context, language, and the agent’s accumulated interpretation of history.

At the same time, LLM agents are not a panacea. Their behavior is stochastic, sometimes inconsistent, and difficult to interpret at the individual level. The value of the framework lies in the *aggregate* phenomena that emerge from populations of LLM agents—bubbles, crashes, path dependence, non-monotonic mixing effects—and in the ability to analyze these phenomena using the tools of nonlinear dynamical systems.

6.2 Individual Rationality vs. Systemic Stability

One of the most thought-provoking implications of our results concerns the relationship between individual cognitive sophistication and systemic stability. Rational agents, who are individually optimal, produce a stable and efficient market. Q-learning agents, who are less sophisticated but follow a consistent learning algorithm, produce moderate instability. LLM agents, who are in some sense the most cognitively sophisticated (they can reason, narrate, and adapt), produce the most instability.

This apparent paradox—more sophisticated agents can lead to less stable markets—is resolved by recognizing that sophistication is not the same as rationality. LLM agents are sophisticated but not rational: they extrapolate trends, construct narratives, and coordinate on shared beliefs, all of which are individually reasonable but collectively destabilizing. This echoes a long-standing insight in behavioral finance and complexity economics: markets are not stabilized by the intelligence of their participants but by the diversity of their expectations (Hong & Page, 2004; Page, 2007). When agents become more similar in their cognitive processes—as they might if all use the same AI tools—market fragility may increase.

This has a direct implication for the real-world adoption of AI in financial markets: if all traders use similar AI models that identify similar patterns and generate similar recommendations, the resulting correlation in trading behavior could amplify rather than

dampen market volatility. The “AI makes agents smarter” narrative may need to be qualified by “AI makes agents more similar, and similarity can be destabilizing.”

6.3 The Role of Heterogeneity

Heterogeneity plays a dual role in our framework. On one hand, agent heterogeneity (different personas, different learning mechanisms) is essential for generating realistic market dynamics: without it, the market either converges to fundamentals (all rational) or exhibits uniform but less interesting dynamics (all Q-learning). On the other hand, excessive heterogeneity in cognitive processes can generate noise and unpredictability that obscures systematic patterns.

Our mixed-population results suggest that the relationship between heterogeneity and stability is non-monotonic and depends on the *type* of heterogeneity. Mixing rational and LLM agents is destabilizing at low LLM shares because the LLM agents introduce noise that the rational agents cannot fully arbitrage. But within the LLM population, heterogeneity in persona (some fundamentalist, some momentum) is stabilizing because the fundamentalist agents provide contrarian pressure. Thus, the effect of heterogeneity depends on whether it introduces diverse trading strategies (stabilizing) or correlated noise trading (destabilizing).

6.4 Does AI Change Economics? A Careful Answer

We return to the question that motivated this paper, now equipped with our results. Does AI change economics?

It does not change economics in the sense of introducing dynamics—dynamics have been central to economics for at least a century, from the cobweb theorem to RBC to ACE. It does not change economics in the sense of replacing rational choice theory—rational choice remains the benchmark and the limiting case.

What AI changes is the *frontier of tractability*. Before AI, studying an economy populated by cognitively sophisticated, narratively motivated, heterogeneously learning agents was either impossible (no way to specify the decision rules) or prohibitively expensive (hand-coding every heuristic). With AI, such economies can be simulated at modest cost, and their dynamics can be analyzed using the standard tools of nonlinear dynamical systems—fixed points, Jacobians, bifurcations, attractors.

This is an incremental expansion of the dynamic economics research program, not a revolution. It makes feasible a class of models that were previously infeasible, and it reveals phenomena—narrative-driven bubbles, path-dependent regimes, non-monotonic mixing effects—that are difficult to generate in traditional models. Whether these phenomena prove important for understanding real economies is an empirical question that this framework helps make tractable.

We therefore conclude that AI’s contribution to economics is best understood as a *methodological expansion*: it extends the set of dynamic economic systems that economists can rigorously study, rather than overturning the existing paradigm. The “paradigm shift,” if there is one, consists in the gradual accumulation of evidence that narratively motivated, adaptively learning agents generate dynamics that are both more empirically realistic and more policy-relevant than those of representative rational agents—and that AI makes this evidence possible to generate.

7 Limitations and Future Work

7.1 Limitations

Our study has several limitations that should be acknowledged.

Small market and simplified asset. Our market has 100 agents trading a single asset with i.i.d. dividends. Real markets have thousands of agents, multiple assets, correlated dividends, and complex institutional features. Our results should be interpreted as qualitative insights rather than quantitative predictions.

LLM cognition as a black box. While we can observe LLM agents’ reasoning traces, we cannot fully control or interpret their internal decision processes. This makes it difficult to isolate the exact mechanisms driving bubble formation and to ensure that the results are not artifacts of prompt design or model quirks.

Cost and scalability. Running 500-period simulations with 100 LLM agents takes hours and costs several dollars per simulation. This limits the number of simulations and the parameter space that can be explored. Larger-scale studies (1000+ agents, 10000+ periods) are currently prohibitively expensive.

No formal estimation. We calibrate rather than estimate the model. Formal estimation—e.g., by matching moments of asset price data or survey expectations—would require substantial methodological development and is left for future work.

External validity. The extent to which LLM agent behavior mirrors human behavior is an open question. While LLM agents reproduce many documented behavioral regularities, they may also exhibit artifacts that do not correspond to human behavior. Our conclusions are conditional on the assumption that LLM agents are a reasonable proxy for human economic cognition, at least at the aggregate level.

Q-learning linear approximation. Our theoretical stability analysis approximates Q-learning agents as linear fundamentalist traders (Eq. (12)). In practice, Q-learning agents make discrete choices via softmax and their demand function is nonlinear; during the learning phase, they may also exhibit momentum behavior because recent high returns reinforce the same action. This approximation may overstate the stabilizing role of Q-learning agents and could affect the accuracy of the theoretical stability conditions. A more rigorous analysis would derive the Q-learning demand function from first principles or estimate it directly from simulation data.

7.2 Future Work

We see several promising directions:

1. **Multi-asset markets:** Extend the framework to multiple assets with correlated fundamentals, enabling the study of contagion, portfolio diversification, and cross-asset bubbles.
2. **Endogenous information acquisition:** Allow agents to choose how much information to acquire (at a cost), rather than receiving a fixed observation set. This would connect to the rational inattention literature (Sims, 2003; Maćkowiak & Wiederholt, 2009).
3. **AI-augmented humans:** Model agents as “human + AI” hybrids, where the AI provides recommendations that the human may or may not follow. This would enable

the study of how AI adoption affects market dynamics when agents have varying levels of trust in AI.

4. **Policy experiments:** Study how regulatory policies (transaction taxes, short-selling constraints, disclosure requirements) affect LLM-agent market stability. This is a natural application given the framework’s ability to model agent adaptation to policy changes.
5. **Comparison with experimental data:** Systematically compare LLM-agent market outcomes with laboratory experimental data (Smith et al., 1988; Haruvy & Noussair, 2006; Hommes et al., 2008) to assess the external validity of LLM agents.
6. **Scalable inference:** Develop methods for estimating LLM-agent models from data, perhaps using simulation-based inference (Cranmer et al., 2020) or moment-matching techniques.
7. **Open-source framework:** Release the codebase as an open-source platform for AI-agent economic simulation, enabling other researchers to design their own markets, agent types, and experiments.

8 Conclusion

We have studied how the learning mechanisms of economic agents shape market dynamics and equilibrium stability in an artificial asset market. By comparing rational, Q-learning, and LLM agents within a unified nonlinear dynamical systems framework, we have shown that:

1. LLM-agent markets exhibit substantially larger price deviations, higher volatility, and more frequent bubble-crash episodes than markets populated by rational or Q-learning agents.
2. The destabilizing effect of LLM agents is driven primarily by momentum extrapolation and narrative coordination, and the transition from stability to instability occurs at a predictable threshold consistent with our bifurcation analysis.
3. In mixed populations, a small share of LLM agents (10–20%) is sufficient to destabilize an otherwise stable market; there is also tentative evidence of a non-monotonic pattern in which very high LLM shares partially restore stability through heterogeneous contrarian behavior, though this latter effect is small relative to simulation noise and requires further investigation.
4. LLM-agent markets exhibit strong path dependence: identical fundamentals can produce divergent long-run outcomes (fundamental convergence, bubble regimes, or depressed regimes) depending on the initial perturbation.
5. The Q-learning rate exhibits a U-shaped relationship with market efficiency, with a flip bifurcation at high learning rates consistent with theoretical predictions.

These findings contribute to the growing literature on AI agents in economic simulation by providing a formal dynamical-systems foundation, systematic benchmark comparisons, and novel results on mixed populations and path dependence. More broadly, they

demonstrate that AI does not “introduce dynamics” to economics—a claim that would be false given the long history of dynamic economics—but rather extends the frontier of what kinds of dynamic economic systems are tractable to study. By making cognitively sophisticated, heterogeneously learning agents computationally feasible, AI opens new avenues for understanding the complex, path-dependent reality of actual markets.

The economy has always been a dynamical system. What AI offers is a more powerful lens through which to study it.

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A LLM Agent Prompt Templates

A.1 System Prompt

You are an investor trading a single asset in a simulated market. The asset pays a dividend each period. You can buy (action = +1), hold (action = 0), or sell (action = -1) one unit each period.

Your investor profile:

- Risk aversion: {risk_aversion} (0 = risk-loving, 1 = very risk-averse)
- Confidence: {confidence} (0 = cautious, 1 = aggressive)
- Fundamental focus: {fundamental_focus} (probability of referencing fundamental value in your reasoning)
- Momentum focus: {momentum_focus} (probability of referencing price trends in your reasoning)
- Narrative tendency: {narrative_tendency} (0 = data-driven, 1 = story-driven)

You will receive market data each period, including the current price, recent price history, dividends, and your portfolio state. Use this information to reason about the market and decide your action.

Learn from your past experience. Your memory contains your previous observations, reasoning, and outcomes.

Output format:

REASONING: <your reasoning in 2-4 sentences>

ACTION: <+1, 0, or -1>

A.2 Period Observation Template

=== Period {t} ===

Current price: {price}

Fundamental value estimate: {fundamental}

5-period moving average: {ma5}

20-period moving average: {ma20}

Most recent dividend: {dividend}

Trading volume last period: {volume}

Your portfolio:

- Shares held: {shares}
- Cash: {cash}
- Portfolio value: {portfolio_value}

Recent price history (last 10 periods): {price_history}

Your memory (recent lessons):
{memory_summary}

What is your action for this period?

B Mathematical Derivations

B.1 Derivation of the Jacobian

Starting from the aggregate price map:

$$p_{t+1} = p_t + \gamma N \left[\omega_R \pi_R(p_t) + \omega_Q \phi_Q(p^* - p_t) + \omega_L \left(\phi_L^f(p^* - p_t) + \phi_L^m(p_t - p_{t-1}) \right) \right]. \quad (28)$$

Define $G_1(p_t, p_{t-1}) = p_{t+1}$ and $G_2(p_t, p_{t-1}) = p_t$. The Jacobian is:

$$J = \begin{pmatrix} \frac{\partial G_1}{\partial p_t} & \frac{\partial G_1}{\partial p_{t-1}} \\ \frac{\partial G_2}{\partial p_t} & \frac{\partial G_2}{\partial p_{t-1}} \end{pmatrix} = \begin{pmatrix} 1 + \gamma N \left[\omega_R \pi'_R(p_t) - \omega_Q \phi_Q - \omega_L \phi_L^f + \omega_L \phi_L^m \right] & -\gamma N \omega_L \phi_L^m \\ 1 & 0 \end{pmatrix}. \quad (29)$$

At the fixed point (p^*, p^*) with $\pi'_R(p^*) = 0$, this reduces to equation (18) in the main text.

B.2 Jury Stability Conditions

For the characteristic equation $\lambda^2 + a_1\lambda + a_0 = 0$, the Jury conditions for both roots to lie inside the unit circle are:

$$1 + a_1 + a_0 > 0, \quad (30)$$

$$1 - a_1 + a_0 > 0, \quad (31)$$

$$1 - a_0 > 0. \quad (32)$$

With $a_1 = -(1 + A)$ and $a_0 = B$ where $B = \gamma N \omega_L \phi_L^m$:

$$1 - (1 + A) + B = B - A > 0, \quad (33)$$

$$1 + (1 + A) + B = 2 + A + B > 0, \quad (34)$$

$$1 - B > 0. \quad (35)$$

Substituting $A = \gamma N [-(\omega_Q \phi_Q + \omega_L \phi_L^f) + \omega_L \phi_L^m]$ and $B = \gamma N \omega_L \phi_L^m$:

$$\begin{aligned} B - A &= \gamma N \omega_L \phi_L^m - \gamma N [-(\omega_Q \phi_Q + \omega_L \phi_L^f) + \omega_L \phi_L^m] \\ &= \gamma N (\omega_Q \phi_Q + \omega_L \phi_L^f) > 0, \end{aligned} \quad (36)$$

which is condition (25). The other conditions follow similarly.

B.3 Flip Bifurcation Condition

A flip (period-doubling) bifurcation occurs when one eigenvalue crosses -1 , i.e., when $\lambda = -1$ is a root of the characteristic equation. Substituting $\lambda = -1$:

$$1 - a_1 + a_0 = 0. \quad (37)$$

With $a_1 = -(1 + A)$ and $a_0 = B$:

$$2 + A + B = 0. \quad (38)$$

Substituting:

$$2 + \gamma N [-(\omega_Q \phi_Q + \omega_L \phi_L^f) + \omega_L \phi_L^m] + \gamma N \omega_L \phi_L^m = 0, \quad (39)$$

which simplifies to:

$$2 + \gamma N [-(\omega_Q \phi_Q + \omega_L \phi_L^f) + 2\omega_L \phi_L^m] = 0. \quad (40)$$

This defines the critical value of the momentum sensitivity (or any other parameter) at which the flip bifurcation occurs.